

Program: Diploma in Fire Technology and Safety	
Course Code: 4467	Course Title: Fire Safety and Fighting lab
Semester: 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To study fire safety properties of hydrocarbons and building materials.
- To evaluate fire suppression agents and extinguishers.
- To conduct performance tests on fire safety equipment

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	CognitiveLevel
CO1	Determine flash point, fire point, and pour point of hydrocarbons.	10	Applying
CO2	Analyze temperature effects on viscosity and material strength.	11	Understanding
CO3	Evaluate fire suppression agents and extinguishers.	10	Applying
CO4	Assess combustibility and flame spread rate of materials	10	Analyzing
	Lab Exam	4	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						1
CO2			2				1
CO3				2			1
CO4							3

3-Strongly mapped, 2-Moderately mapped, 1-Weaklymapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	CognitiveLevel
CO1	Determine flash point, fire point, and pour point of hydrocarbons.		
M1.01	Flash point and fire point determination	5	Applying
M1.02	Effect of temperature on viscosity using Redwood viscometer	5	Applying
CO2	Analyze temperature effects on viscosity and material strength.		
M2.01	Strength test on concrete and masonry units	5	Understanding
M2.02	Rebound hammer test on heated concrete	6	Understanding
	Lab Exam I	2	
CO3	Evaluate fire suppression agents and extinguishers.		
M3.01	Testing of Dry Chemical Powder (DCP)	5	Applying
M3.02	Testing of AFFF (Aqueous Film Forming Foam)	5	Applying
CO4	Assess combustibility and flame spread of materials		
M4.01	Flame spread rate determination	5	Analyzing
M4.02	Non-combustibility tests on building materials	5	Analyzing
	LabExam–II	2	